MILITARY SPECIFICATION

FITTINGS, FLARELESS TUBE, FLUID CONNECTION

This amendment forms part of Military Specification MIL-F-18280D, 4 Feb. 1980, and is mandatory for use by all Department and Agencies of the Department of Defense.

PAGE 5

a. Insert "AS 1376 Wrench Pads for Fluid Fittings Machined from Alternate Shapes of Material."

(Copies of the above publication may be obtained from Aerospace Industries Association of America, 1725 DeSales Street, N.W., Washington, DC 20036.)"

PAGE 6

Para. 3.2.1.1 after "defects" insert "exceeding 0.010 inch deep". After MIL-H-6868 add sentence: "Surface imperfections due to tool or die marks of 0.010 inch or less are acceptable."

PAGE 7

Table I:

a. Delete "C" in front of all materials.

b. Delete all flag 3/'s from table.

c. Delete "3/ Nonleaded type only" from notes at bottom of table.

d. Insert "if applicable" after AS 1376 at bottom of page.

e. In "Material Code" column.

1. Opposite "Forgings" delete "None" and insert letter "F".

2. Opposite "Bars" delete letter "B" and insert "None".

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:11Z
Check the source to verify that this is the current version before use.

f. In "Type of Part" column.

1. Delete "Forged" and insert "Shape" in 2nd block and "Shape and" in 3rd, 4th and 5th blocks.

PAGE 10

Para. 3.5.1

a. Delete "the letter B and" from 2nd sentence.

Para. 3.5.2

a. At end of Para. insert "Carbon steel forgings made from 4130 shall be marked with the letter F".

PAGE 11

Para. 3.6 Change "3.6.6" to "3.6.5" and "3.6.7" to "3.6.6".

PAGE 13

Para. 3.6.6

a. Delete and insert new paragraph as follows.

"3.6.6 Joint strength. Flareless tube fitting assemblies shall be tensile tested per 4.7.12. The assemblies shall withstand, as a minimum, the loads as specified in Table V without slippage of the sleeve on the tubing or cracking of the sleeve. Sleeve failure shall not be attributed to other assembly components. Fitting assemblies shall be per figure 6, carbon steel on MIL-T-6845 tubing. Joint strength test qualification and quality conformance tests are not required for -24 and -32 size sleeves."

Table V.

a. Delete and insert new table as follows.

SIZE	-2	-3	-4	-5	-6	-8	-10	-12	-16	-20
Wall Thickness (Inches)	.012	.016	.020	.020	.028	.035	.042	.058	.065	.049
Lbs.	400	800	1500	1800	2500	4200	6200	8800	10000	9500

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:11Z
Check the source to verify that this is the current version before use.

Para. 4.2.2 after "analysis" insert "magnetic particle inspection (3.2.1.1)".

Para. 4.3

a. Delete "Retention of qualification" paragraph and insert new paragraph as follows.

"Retention. The retention of qualification shall consist of certification by the manufacturer to demonstrate compliance of the qualified fittings with the requirements of this specification. Certification shall be signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number or designation; and meets the requirements of the current issue of the specification. Certification shall be conducted at intervals not exceeding 2 years. The activity responsible for certification approval is the qualifying activity (see 6.3)."

PAGES 15 and 16

Para. 4.4.2

a. Change 4.7.8 to 4.7.7, 4.7.7 to 4.7.6, 4.7.11 to 4.7.10, 4.7.12 to 4.7.11 (after 3.6.6), 3.6.6 to 3.6.5, 3.6.7 to 3.6.6.

b. Insert "(Sleeves only)" after "Joint Strength Test".

Para. 4.5.1.2

a. Delete "(f)".

Para. 4.5.2

a. Delete 3.2.2.2.3.

b. Change 4.7.6 to 4.7.5, 4.7.10 to 4.7.9, 4.7.11 to 4.7.10, 4.7.14 to 4.7.13.

c. Insert ("Sleeves only)" after "Joint Strength Test".

d. Delete "(K) Carbon steel forged fittings torque test (3.2.3, 4.7.15)."

e. Insert "(J) Bar stock magnetic particle inspection (MS21921 nuts only) (3.2.1.1, 4.2.2)."

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:11Z
Check the source to verify that this is the current version before use.

f. Delete "Note paragraph" and insert new "Note paragraph" as follows.

*Note: For quality conformance testing, the joint strength tests shall be limited to 3 assemblies using 304 1/8 hard tubing to MIL-T-6845, MS 21922 sleeves and carbon steel straight fittings per figure 6. Six sleeves shall be tested.

PAGE 17

Table VI.

a. Change 4.7.7 to 4.6.6, 4.7.8 to 4.7.7, 21902 (opposite "Joint strength test") to 21922.

b. Change Note after flag 5/ at bottom of table to read "To be tested with three straight carbon steel fittings on 304 1/8 hard tubing to MIL-T-6845. Six sleeves shall be tested."

PAGE 19

Para. 4.7.3.1

a. After "ARP 603", Insert "except temperature shall be at 100 $\pm$ 30°F.

Para. 4.7.4 1st sentence after "tees" insert "MS 21902 unions (repeated assembly samples only)".

PAGE 20 and 21

Para. 4.7.6

a. 1st paragraph, delete last sentence and insert new sentence as follows: "The bending stress must be measured by one or more strain gages placed on the tube 3/16 $\pm$ 1/32 inch from the tail of sleeve."

b. After "28,000,000" insert "P.S.I.".

PAGE 21

Para. 4.7.6.1

a. Delete 2nd sentence.

b. Delete "apparent" from notes below Table X.

PAGE 22

Para. 4.7.6.2 After "28,000,000" insert "P.S.I.".

PAGE 23

Para. 4.7.6.3 After "28,000,000" insert "P.S.I.".

Source: https://assist.dla.mil -- Downloaded: 202640-18T11:11Z
Check the source to verify that this is the currentversion before use.

Para. 4.7.7.1

Last sentence before "assembly" insert "MS21902 fitting".

Para. 4.7.7.2

a. 1st line change 4.7.8.1 to 4.7.7.1.

PAGE 24

Para. 4.7.11

a. 3rd sentence after "air or" delete "N2" and insert "nitrogen", add period and start next sentence by starting the word samples with a capital "S". Replace the word "or" with "and" after the word "clean".

Para. 4.7.12

a. 1st line change "six" to "three".

b. At end of paragraph add "Presetting of sleeves may be done by hand or machine (for qualification and quality conformance testing of MS 21922 sleeves only)".

PAGE 32

Not numbered,

a. Add page no. "32".

b. Delete Figure 4 and insert new Figure 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:11Z
Check the source to verify that this is the current version before use.

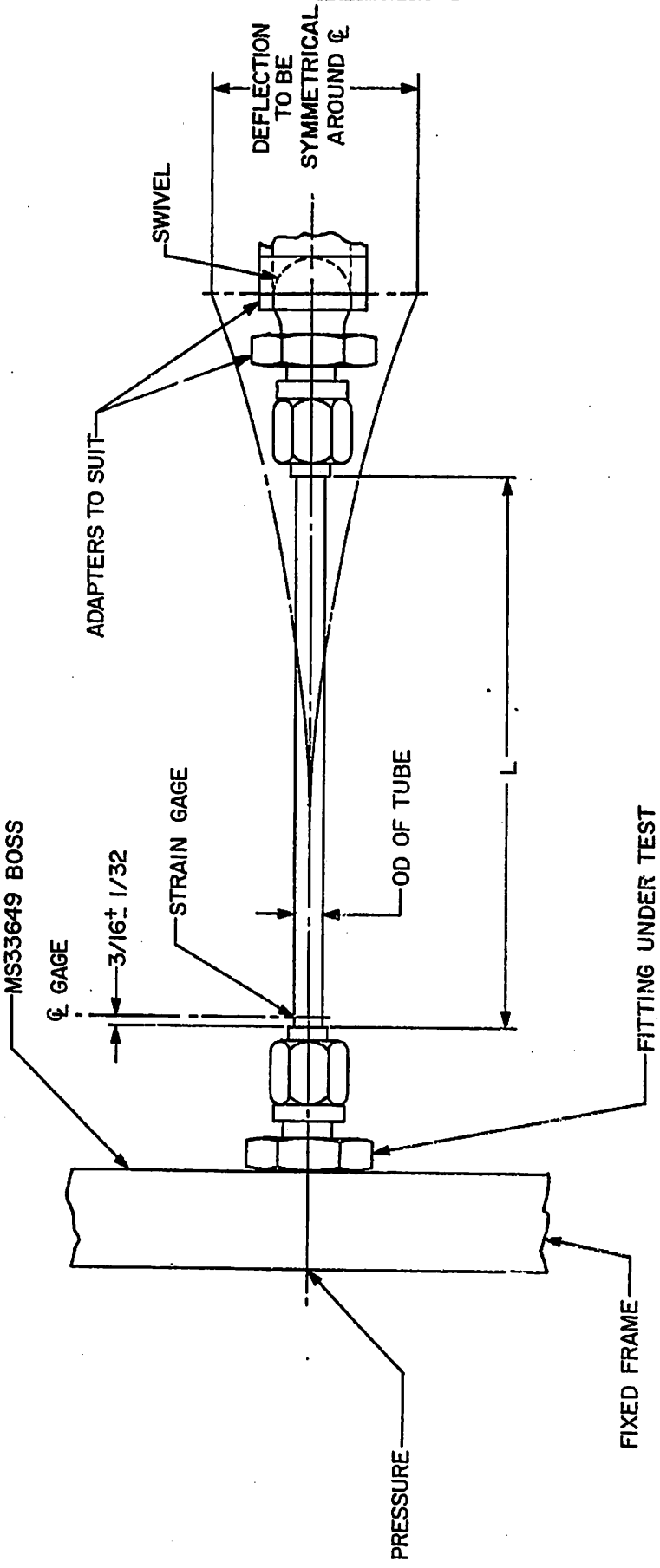


Figure 4. Rotary beam method.

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:11Z
Check the source to verify that this is the current version before use.

PAGE 33

Figure 5

a. Delete "(4 places)".

PAGE 34

Figure 6

a. Add following note.

"Alternate test configurations are allowed, such as attaching fitting to tensile test machine instead of tubing. However test conditions of paragraph 4.7.12 must be met."

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing Activity:
Navy - AS
Project No. 4730-0145

Review:

Army - AR, MI
Air Force - 11
DLA - CS

Source: https://assist.dla.mil -- Downloaded: 202640-18T11:11Z
Check the source to verify that this is the current version before use.